

Human-centric Intelligibility for Automated Vehicles: Aligning Explanations with Predictive Models of Passenger State

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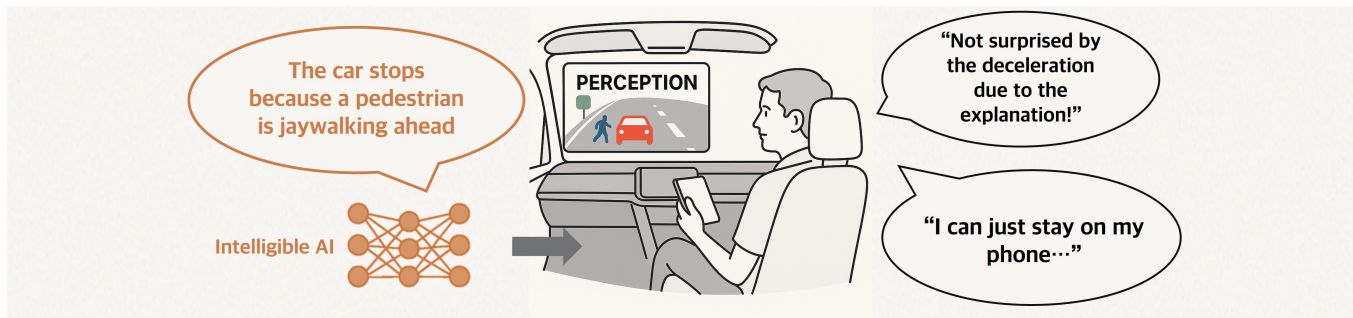


Figure 1: Intelligible explanations enhance passenger experience in AVs by providing contextually relevant information at the precise moment it's needed.

Abstract

Public acceptance of automated vehicles (AVs) is hindered by a lack of trust due to their unintelligible decision-making. This research bridges the gap between explainable AI development and passenger experience by investigating human-centric explanation systems for AVs. We developed and validated an on-road Wizard-of-Oz evaluation platform to test AI-generated explanations safely. Through this platform, we created a comprehensive multimodal dataset capturing passengers' in-situ explanation demands during real driving scenarios, including physiological responses, vehicle dynamics, and environmental context. Our findings reveal that passengers need to understand what the vehicle is doing rather than why, with perception-based visualizations proving more effective than abstract attention heatmaps for building trust and situational awareness. We demonstrate that a predictive model using vehicle dynamics and physiological states can identify opportune moments for explanation delivery. As we extend towards generalizable modeling, this research aims to provide empirical guidelines for intelligible AVs and establish a framework for intelligible in-vehicle systems that dynamically align explanations with general and individual passengers' informational needs, fostering more adaptive and trustworthy human-vehicle interactions.

CCS Concepts

• Human-centered computing → Ubiquitous computing.



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Keywords

Automated vehicles, explainability, intelligibility, automotive user interfaces

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1 Problem Statement

Recent development of highly automated vehicles (SAE Levels 4-5) is enabling passengers to disengage from driving tasks and perform non-driving related tasks (NDRTs) to spend the journey productively and entertainingly (e.g., watching VR media in cars [5–9, 12, 37, 38]). However, widespread adoption faces challenges related to user anxiety and trust deficits in automated systems [11, 25], particularly when vehicles exhibit unexpected behaviors [30]. Research indicates that explanations of vehicle actions and intentions can improve passenger control perception and trust calibration [26]. Thus, alongside efforts to improve the robustness of automated driving itself [16], similar emphasis is placed on how we should design the interactions with these systems to be more interpretable and inclusive [18] (i.e., how AVs can explain their perceptions and decisions to passengers [2] and other road users [13, 14]). However, a research gap persists between algorithmic practices for explainable artificial intelligence (XAI) development and passenger requirements in operational contexts.

Existing XAI models for automated driving predominantly target AI and data expert developers. This orientation creates a disconnect wherein the timing of explanations strongly influences passenger

trust, yet machine learning research emphasizes content generation over temporal delivery considerations [15]. Consequently, mistimed or superfluous explanations may compromise user experience [15]. This limitation stems from the insufficient multimodal datasets that document real-time passenger explanation needs and the lack of ecologically valid testing methodologies for in-vehicle explanations.

Additionally, the engagement of passengers in NDRTs within AVs introduces complexity to explanation system design. Effective explanations must demonstrate accuracy and timeliness while avoiding disruption to passengers' primary activities, which may otherwise result in frustration or cognitive overload. Inappropriate timing or excessive complexity in explanations can prove counter-productive. In an extreme case, when a passenger has just woken from sleep, their visual and cognitive processing capabilities are limited, making it impossible to comprehend a complex driving scene at a glance [23]. In such moments, a generic explanation is insufficient; a tailored, easily digestible one is required. This research therefore extends beyond traditional driver-centric interface design to address the distinct interaction requirements of passengers, for whom attention competes with NDRTs rather than driving tasks and whose situational awareness level is substantially lower than drivers. The problem involves not merely achieving AI explainability, but ensuring explanation intelligibility and contextual relevance for passengers.

This research investigates the design and evaluation of passenger-centric explanation systems for AVs (Figure 1). The primary objective is to establish connections between AI-generated explanations and passenger user experience through examination of the following research questions:

- (1) What types of explanations do passengers require during automated drive, and when do these demands arise?
- (2) How do variations in explanation types and timing affect passenger trust, perceived safety, situational awareness, and cognitive load?
- (3) How can empirical findings contribute to a technology acceptance model for explainable AVs?

Through addressing these questions, this research aims to establish foundational principles for developing in-vehicle explanation systems that help calibrate human trust towards AVs.

2 Related Work

Explanations serve an essential function in calibrating passenger trust to correspond with the vehicle's actual capabilities, thereby mitigating both the risks associated with dangerous overtrust and the limitations imposed by hesitant undertrust [3]. The phenomenon of overtrust warrants particular attention, as it may engender complacency or foster a misplaced sense of security. This concern persists even in SAE Level 4 vehicles, where passengers are not required to intervene but nonetheless must comprehend the system's operational domain [15]. Conversely, undertrust may result in the rejection of potentially beneficial technological innovations. Explanations that are inadequately designed or inappropriately timed may not be delivered as intended to passengers, potentially failing to address these concerns and thereby compromising the passenger experience [10, 19].

Modern developments in XAI have yielded models capable of describing vehicular decision-making processes through visual or textual representations. These approaches frequently emphasize the visualization of machine "attention" (exemplified by saliency maps) or "perception" (illustrated through arrows) [21, 32]. Nevertheless, such algorithm-centric explanations remain insufficiently evaluated with human passengers in naturalistic contexts, resulting in limited understanding of their practical implications for user experience. This constitutes a methodological limitation, as explanations that demonstrate technical accuracy may have limited intuitive comprehensibility for non-expert passengers. Moreover, passengers' capacity to engage in NDRTs introduces additional complexities [1], as explanations must compete for finite attentional resources, thereby rendering the temporal and modal aspects of information delivery a confounding problem [15, 19].

One hurdle to the development of passenger-centric explanations resides in the limited availability of publicly accessible, ecologically valid datasets that annotate how human passengers behave and perceive. The majority of current datasets related to explainable driving are derived from recorded driving videos and lack provisions for capturing the passenger's contextual state, including their informational requirements. To address this methodological gap, this investigation employs the Wizard-of-Oz (WoZ) methodology [4], an established experimental paradigm that facilitates the safe examination of prospective automated systems. This approach circumvents ethical and regulatory concerns while enabling realistic automated driving environments, thereby allowing the collection of ecologically valid data concerning passenger behavior and preferences in AVs. By elucidating passenger requirements, this approach extends the theoretical framework of in-vehicle "interruptibility," which has traditionally emphasized driver-centric considerations. Within the paradigm of fully automated vehicles, this conceptual framework undergoes a fundamental transformation, necessitating the identification of optimal moments for "explainability" when addressing passengers engaged in NDRTs [15, 24].

3 Human-centric Intelligibility for AVs

This research employs a three-fold approach to advance our understanding of passenger-centric explanations in AVs. Phase 1 [19, 20] establishes an on-road evaluation platform to determine what types of explanations are most effective and when they should be presented. Phase 2 [15] investigates the alignment between passengers' physiological states and their individual explanation needs, creating a comprehensive multimodal dataset for predictive modeling. Phase 3 envisions broader deployment by modeling explanation demands at scale. This structure enables systematic research: 1) establishing fundamental explanation principles, 2) understanding individual variations in explanation needs, and 3) generalizing these insights for widespread implementation.

3.1 Phase 1: Evaluating Whether and How Explanations Affect Passenger Experience

3.1.1 Methodology. Apparatus and Platform: A WoZ vehicle driven by a wizard driver was equipped with a Varjo VR-2 Head-Mounted Display (HMD) worn by passengers. A 360° camera provided a see-through video feed [35, 36], augmented with visual

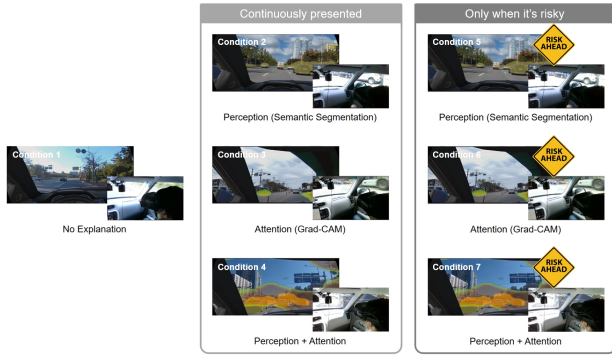


Figure 2: Comparison of types and timing mechanisms of visual explanations [19].

explanations on a virtual Windshield Display (WSD). The HMD transported passengers' visual perception into a virtual world where the wizard driver was removed, creating the illusion of automated driving. The Unity-Python framework enabled registration of the vehicle's real-time predictions on the virtual WSD, where virtual elements were overlaid onto the real-world driving environment.

Explanation Algorithms: The platform integrated real-time AI models for generating visual explanations. Perception was visualized through semantic segmentation maps from PIDNet [31], while vehicle attention was represented as Grad-CAM [27] heatmaps derived from a custom 3D CNN trained for traffic risk prediction.

3.1.2 Evaluation. Experimental Design: A 3 (Explanation Type: Perception vs. Attention vs. Both) $\times$ 2 (Timing: Always-on vs. Risk-adaptive) within-subjects study was conducted with 30 participants. All seven conditions, including a no-explanation baseline, were experienced in a counterbalanced order to mitigate learning effects (Figure 2). To ensure ecological validity and test the explanations under various real-world scenarios, the experiment took place on a predefined network of routes. These routes were designed to be diverse, composed of 2.4km urban roads with a 60 km/h speed limit, 3.1km arterial roads with a 70 km/h limit, and 2.9km local highways with an 80 km/h limit. This variety exposed participants to different traffic densities, speeds, and driving events, including numerous crosswalks and intersections, providing a robust testbed for the explanation systems.

What and When to Explain?: The evaluation revealed that visualizing vehicle perception through segmentation maps significantly enhanced perceived usability, trust, safety, and situational awareness. Attention heatmaps were perceived as indirect and requiring interpretation, increasing cognitive load. While continuous explanations were generally preferred, risk-adaptive presentation effectively improved trust and perceived safety when displaying information-rich content (Perception + Attention). Physiological data corroborated this finding, with significantly lower phasic EDA ($p < .05$) in risk-adaptive conditions versus always-on presentation for combined explanations.

Structural equation modeling revealed that AV acceptance is primarily driven by perceived safety, trust, and vehicle understandability, with perception-based explanations directly influencing these critical factors.

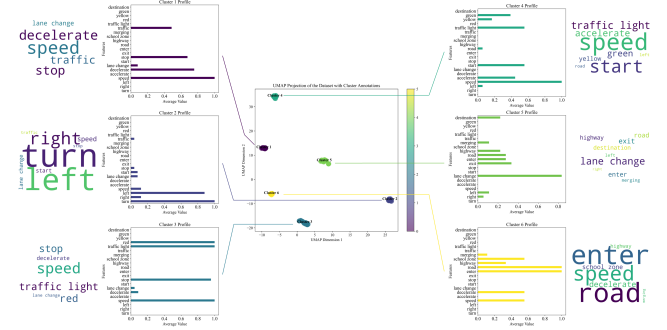


Figure 3: Taxonomy of the required explanations [15].

3.2 Phase 2: Aligning Passengers' States with Individual Explanation Needs

3.2.1 Methodology. Data Collection Protocol: Building on the insights from the initial study, this phase employed a more intensive data collection protocol to model the underlying factors of passenger explanation needs. Using the WoZ vehicle with a partition instead of the HMD and a 14.8km route featuring urban, arterial, and highway segments, the focus shifted from evaluating predefined interfaces to capturing a holistic, multimodal view of the passenger's state. Twenty-nine new participants completed 30-minute drives while freely engaging in NDRTs. They were instructed to explicitly indicate their need for an explanation by stating "now," at which point an experimenter would mark the event. The passenger would then articulate their specific informational need, which was recorded to provide ground-truth labels.

Multimodal Data Acquisition: An expanded sensor suite captured synchronized data streams via UNIX timestamps: ZED stereo camera and VLP-16 3D LiDAR for environmental sensing; GPS, IMU, and OBD-II data for vehicle state; Intel RealSense depth cameras for posture analysis, Lepton 3.5 thermal camera for cognitive load indicators, custom 32 $\times$ 32 pressure-sensing seat for action recognition [28], and Empatica E4 wristband for physiological responses (GSR, BVP, HR) for being indices of situation awareness [17] and arousal [22], using a modified version of the ActionSense framework [29].

3.2.2 Evaluation. Individual Variations in Explanation Needs: Passengers requested explanations averaging 4.5 times per 30-minute drive. K-means clustering revealed six distinct recurring themes (e.g., "decelerating due to traffic," "stopping for red light," "lane change for highway exit"), demonstrating that demands are systematically tied to specific driving contexts (Figure 3). Analysis revealed that 50.7% of requests sought information about vehicle actions without requiring justification. When reasoning was requested, it primarily concerned traffic lights or surrounding vehicles. Notably, no participants requested "why" explanations without corresponding "what" information.

Predictability through Physiological and Motion Data: Machine learning analysis demonstrated that models integrating vehicle motion and passenger physiological data could predict explanation demand timing with 86.4% balanced accuracy using Bi-LSTM-CNN architecture. Permutation importance analysis identified phasic GSR and IBI as the most predictive features, followed by vehicle speed, establishing a direct link between passengers' internal states and information needs.

3.3 Phase 3: Scaling Personalized Explanations for General Deployment

The future phase will synthesize insights from Phases 1 and 2 to develop adaptive explanation systems suitable for diverse passenger populations. One limitation of the sensor-based approach is its tendency to overfit to individual passengers due to the highly personalized nature of explanation demands. This phase aims to develop a generalizable framework that leverages the dataset from Phase 2 to predict explanation needs across different user populations in real-time, without requiring explicit passenger requests.

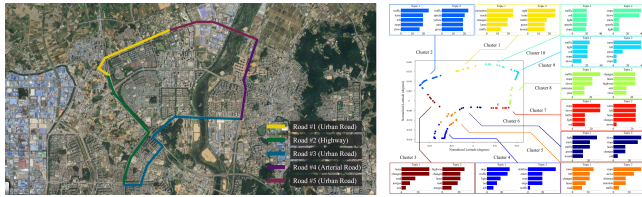


Figure 4: City-wide Modeling of Explanation Demands [15].

3.3.1 City-wide Modeling of Explanation Demands. The dataset from Phase 2, with its rich, geo-tagged explanation demands, opens up the possibility of scaling the analysis from individual routes to entire urban environments [15]. The current methodology, which relies on a highly instrumented vehicle, can be adapted for large-scale, crowdsourced data collection. By leveraging the ubiquitous sensing capabilities of modern smartphones (for GPS and IMU data) and the widespread availability of Bluetooth-enabled OBD-II dongles, a lightweight version of the data collection framework can be deployed to a larger fleet of vehicles. This would enable the creation of city-wide "explanation demand heatmaps", identifying specific intersections, road segments, or merging zones that consistently generate passenger uncertainty or anxiety. The analysis from the initial dataset, which used clustering on GPS data and the TextRank algorithm to identify geographically-specific explanation themes (e.g., lane changes for a particular destination, merging behavior on a specific highway), serves as a proof-of-concept for this approach (Figure 4). Such a city-wide model would inform vehicle manufacturers, urban planners, and traffic engineers seeking to identify and mitigate points of confusion or cognitive load in the driving scenes.

3.3.2 Future Work: Fine-Tuning Vision-Language Models with Passenger State. The two completed studies establish the foundation for developing adaptive explanation systems in AVs. Building on these findings, this section outlines future work that integrates explanation design principles with state-of-the-art AI algorithms

to enable generalizable prediction of both explanation content and demand timing in real-world driving scenarios.

Recent advancements in Large Language Models (LLMs) have shown capabilities in generating descriptive explanations from driving videos [33, 34]. However, these models are trained on driving videos, which lack human annotations for the explanation demand. While they can describe observable events, they cannot infer what passengers need to know or when explanations are necessary.

This research will address this gap by fine-tuning pre-trained VLMs with human annotations for passenger state-related labels into the explanation generation process. By training on synchronized multimodal data that correlates driving situations with explicit explanation requests, the model will learn to generate explanations that are both factually accurate and contextually relevant.

For example, while a generic LLM might output "The vehicle is decelerating", a fine-tuned model would recognize patterns linking sudden deceleration with elevated passenger arousal and generate more informative explanations or suppress explanations when passengers would have shown no signs of concern. This will create systems that are not merely explainable but intelligible, aligning communication strategies with passengers' inferred states. By moving beyond individual-specific models to learn generalizable patterns of explanation demand, this work will establish scalable approaches for building trust in human-vehicle interactions.

4 Contributions and Impact

The key contributions of this research are:

- (1) **A Dataset for Passenger-Centric XAI:** This research contributes a comprehensive, multimodal resource for studying passengers' real-world explanation demands [15]. The dataset provides synchronized environmental, vehicle, and passenger sensing data alongside in-situ explanation needs.
- (2) **An Ecologically Valid On-Road Evaluation Framework:** The mixed-reality WoZ platform bridges the gap between laboratory simulations and real-world deployment, offering a replicable methodology for assessing human-in-the-loop impacts of AI explanations [19].
- (3) **Empirically Grounded Design Guidelines:** Findings from both phases provide actionable design principles, such as prioritizing perception-based visualizations and recognizing that operational transparency ("what") often supersedes causal reasoning ("why").
- (4) **A Predictive Model Linking Driving Situation and Explanation Needs:** The research demonstrates that multimodal sensing can predict explanation demands, paving the way for proactive, personalized explanation systems that adapt to individual passenger requirements.

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